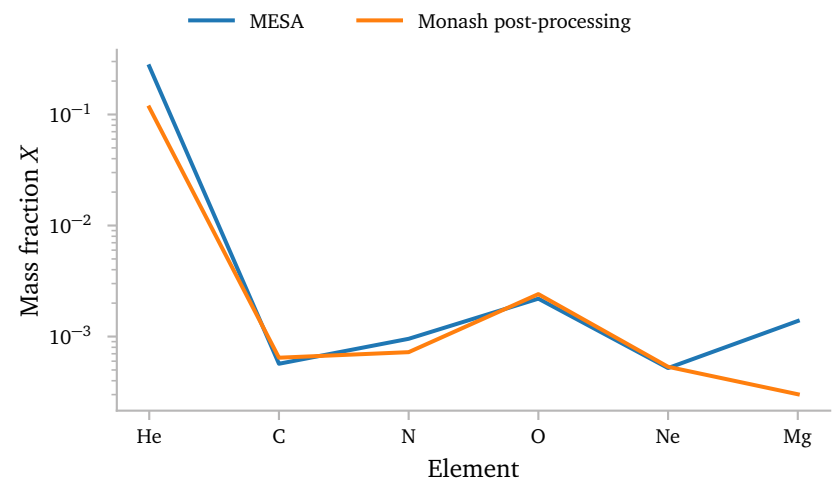


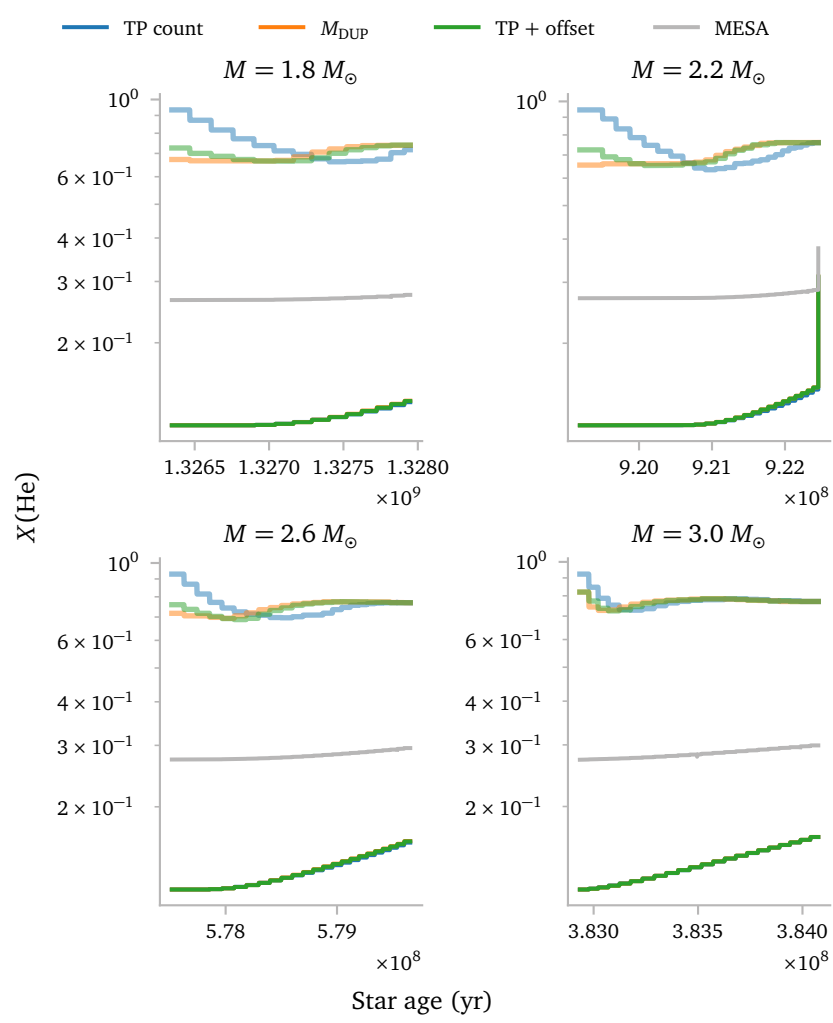
Notes Week 28

a. Monash Initial Helium Envelope Abundances

Last week, I showed these two plots:

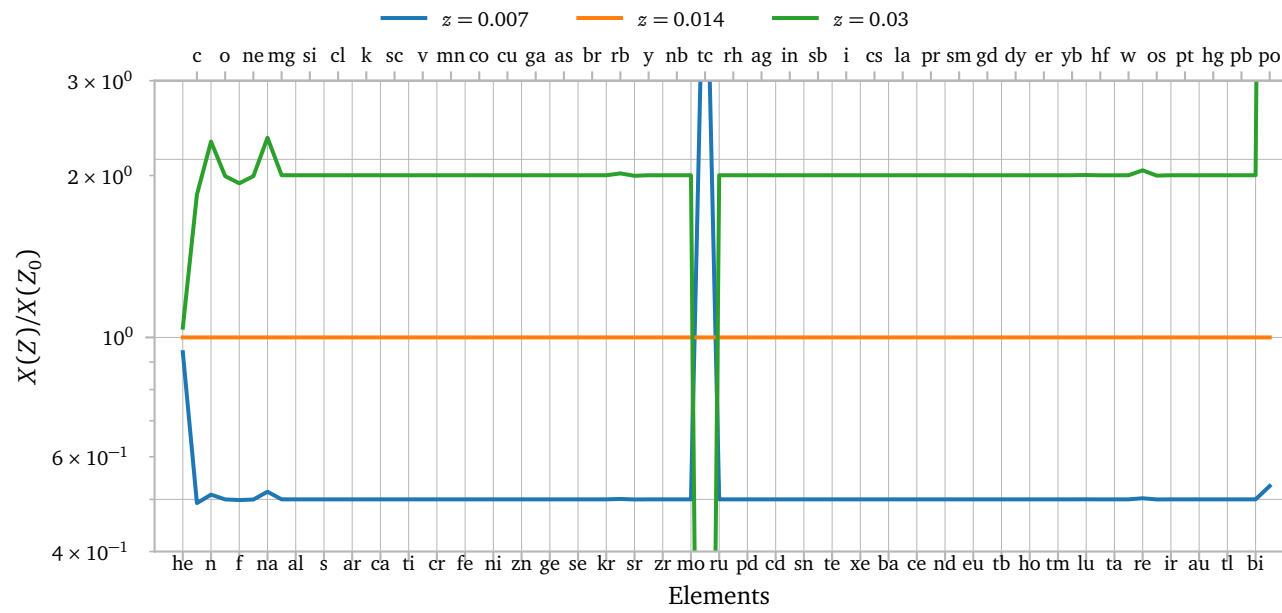


The most important thing we saw here was the quite large differences between the helium mass fractions.



Here, the same difference is again very clear.

It is actually very easy to understand where this discrepancy comes from, and in fact, I have already made a figure before that shows exactly what is going wrong here:

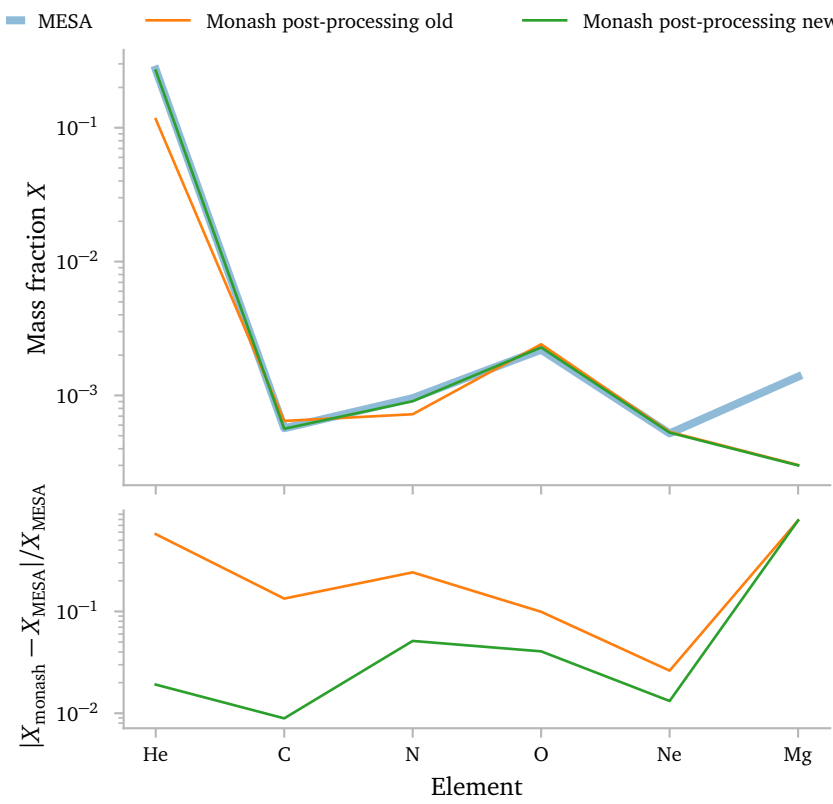


Here I plot the initial envelope abundances for 3 different Monash metallicities divided by the Monash solar Metallicity model.

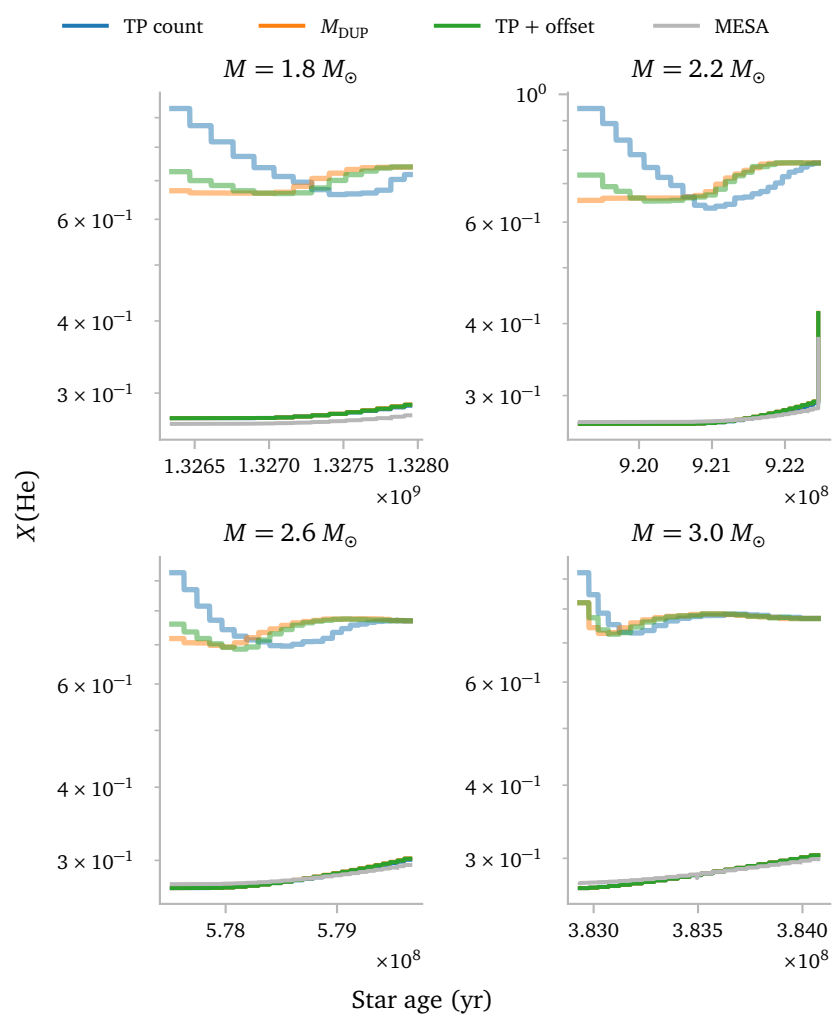
I made this plot to *confirm* I can simply take the abundance of an element, divide it by the Monash metallicity and multiply it by the MESA metallicity to get the MESA envelope abundance.

Now I think this works great for pretty much all elements, *except* for some of the very light elements, exactly the elements that I was comparing before.

I quickly fixed this by also interpolating the initial envelope abundances.



Clearly, this works much better for helium and actually all other isotopes too.



This looks much better!

b. MESA intershell abundances

In order to verify the Monash enrichment is working well, we can compare the enrichment of the envelope as predicted by my Monash post-processing code with the abundances from MESA.

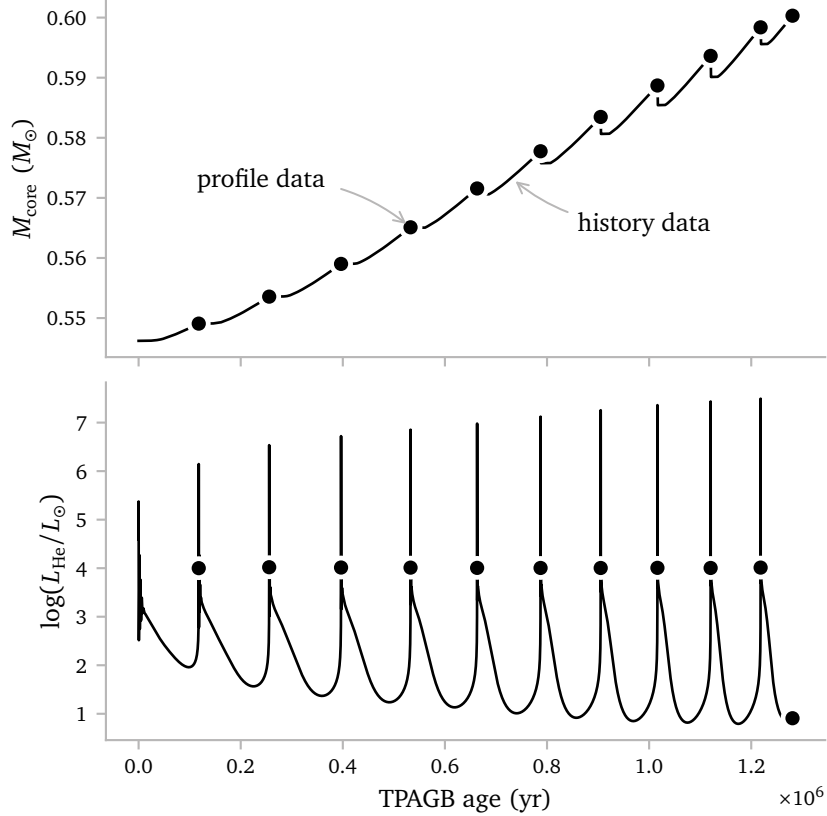
I think there are a couple interesting things to do. First, we just want to compare the intershell abundances of the elements that are available to me in the MESA models with the predicted Monash intershell abundances. This is already interesting, because this gives us a good idea of if the interpolation is working well.

Then, I think it is interesting to look at the envelope abundances to compare the differences. I have already done this previously, *but* now I have a “better” initial envelope abundance, and insight into the intershell abundances of both codes.

Lastly, I want to verify that my envelope enrichment code is working correctly. I will do this by injecting the MESA intershell abundances and the initial envelope abundance into my post-processor. I will then compare the envelope abundances it spits out with the actual MESA envelope abundances. If the post-processor is working well, it should reproduce the MESA envelope abundances well¹.

b.1 Intershell abundances

To start, I document *how* I obtain the intershell abundances using only a few profiles.



¹ Although I do not expect it to agree completely, because the dredge up assumed to be instantaneous, and the precise derived dredged up masses are slightly different from the MESA internal dredged-up masses.

For all single stars, I save a model and profile at the start of a thermal pulse, just before things really start to change and the helium burning is just beginning to start up.

These profiles contain a lot of relevant information, but they are sparse. To see *when* in the lifetime of stars the profiles are saved, I show the core mass and the helium luminosity.

Clearly, the models are correctly saved at the start of a thermal pulse. I think this is quite a nice time to sample the intershell abundance, since this is roughly also the abundance when dredge up occurs.

The last profile is an outlier. This is the model that is saved when the envelope mass is below 0.01 $M_{\odot}$.

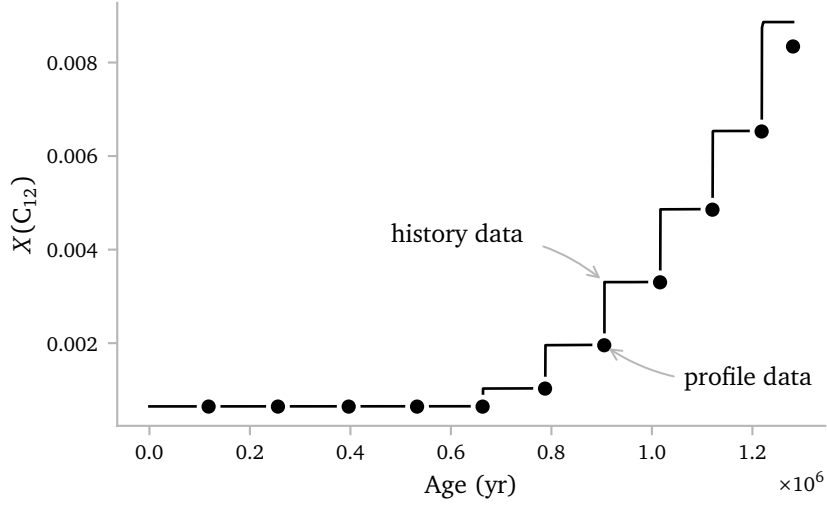
Now let me plot something a bit more interesting. By finding the position of the greatest change in the mass fraction of the metals²,

```
i = np.argmax(
    np.abs(
        np.log10(p.z_mass_fraction_metals[10:])
        - np.log10(p.z_mass_fraction_metals[:-10])
    )
)
```

we get a proxy for the intershell / envelope boundary.

By then simply taking the mass averaged abundance of an element a certain margin before this boundary, we get the envelope abundance. Since we also have my MESA implementation of the envelope abundances, we can compare the two.

² over a small number of zones



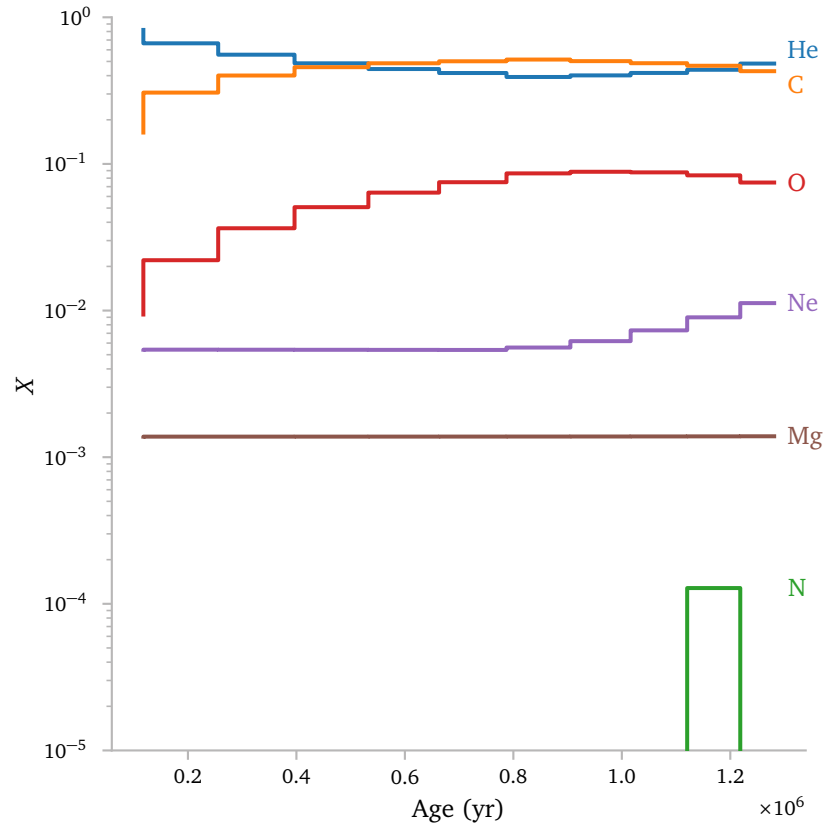
We clearly see that the two agree very well, even though they are computed with slightly different methods.

The MESA history data is based on the largest convective region. It then takes a mass average of this region. Luckily, and also unsurprisingly, this agrees well with the method that simply takes the mass averaged value of the abundances past the intershell zone.

The only difference occurs when the envelope mass is very small.

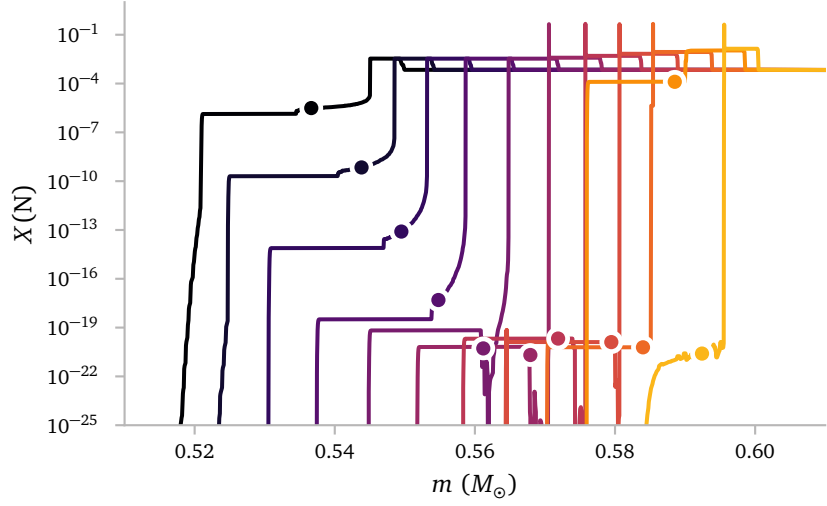
The shape of the envelope abundance, and very likely also the intershell abundance, is very nicely approximated by a step function.

Using the same criterion, but now taking some average to the intershell side of the intershell / envelope boundary, we can also find the intershell abundances. Unfortunately however, we cannot compare this with MESA history data.

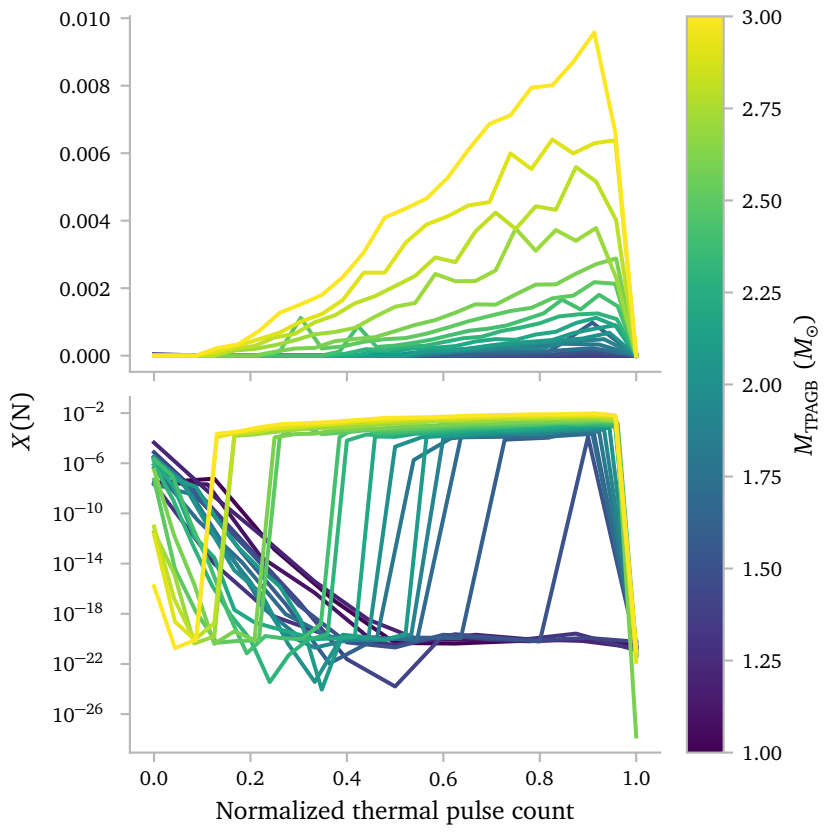


Here we can see all available elements together to see the relative abundances.

Everything looks normal to me, except for weird nitrogen bump. The nitrogen abundance is constantly



This is weird I think. But it is not necessarily something that is wrong with my intershell abundance finding code. There is just something that happens during this specific thermal pulse that makes the intershell relatively very abundant.



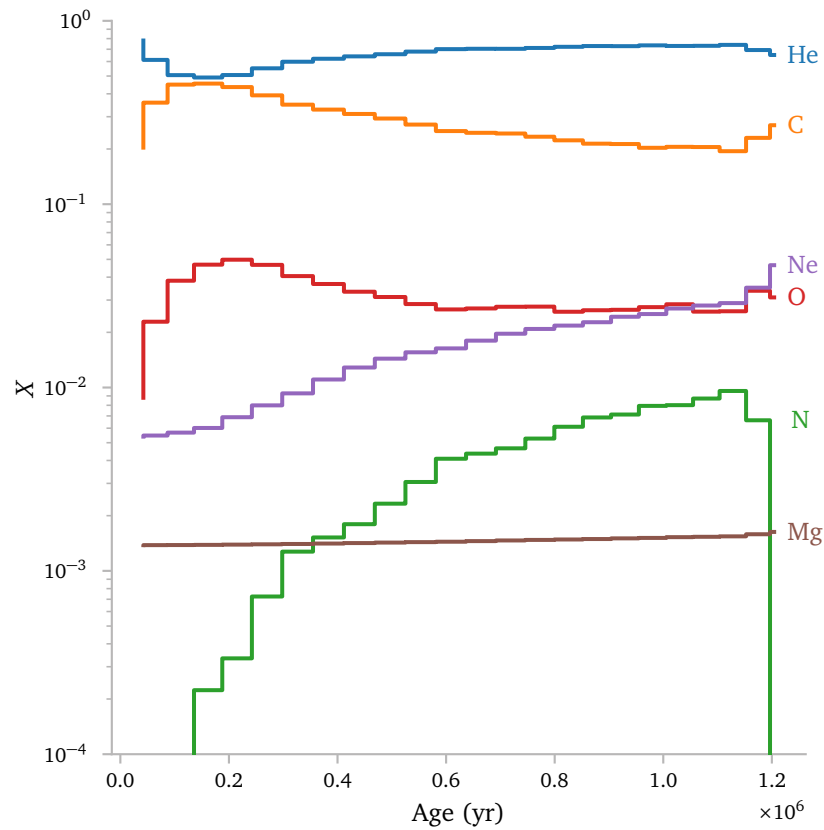
Apparently, the 1.5 $M_{\odot}$ model just by accident, only has 1 thermal pulse with a significant nitrogen abundance. This however does not mean that all stars have this 1 thermal pulse with a large nitrogen abundance, and it also does not seem like an error since there is quite a clear pattern.

Clearly, stars with masses larger than 1.5 $M_{\odot}$ have multiple pulses with a large nitrogen abundance. Stars with masses less than 1.5 $M_{\odot}$ have *no* thermal pulses with a large nitrogen abundance.

I do not know what causes these sudden increases in envelope abundances and why the abundances drop to low values again during the last thermal pulse.

Maybe it has something to do with the CNO cycle?

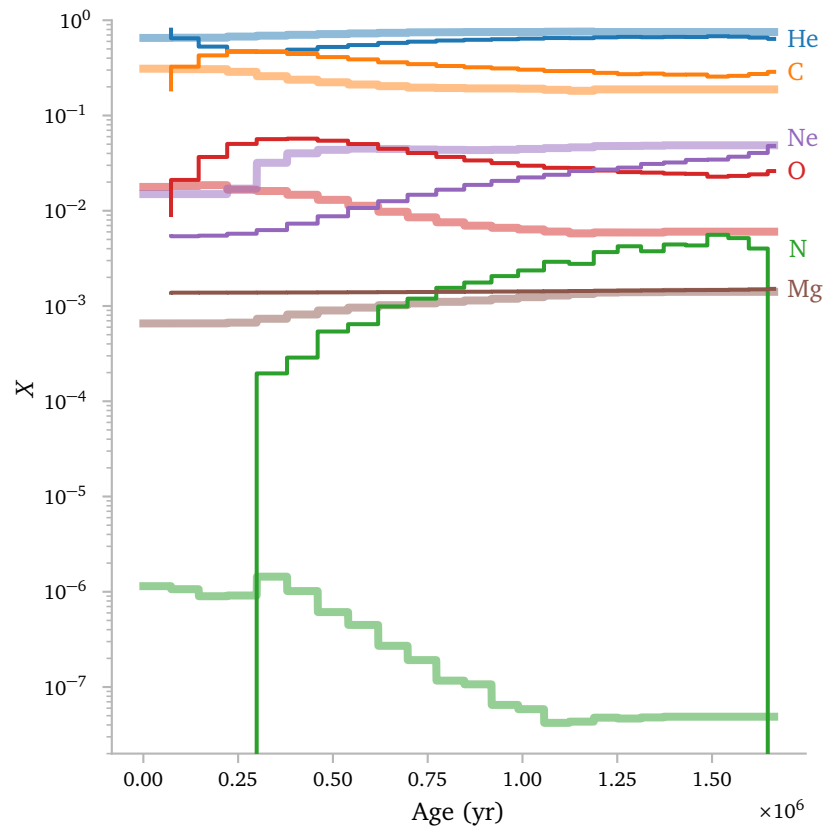
Obviously, the mass is quite important for the nitrogen abundance. Let me also plot the intershell abundances for the 3 $M_{\odot}$ MESA model.



Obviously the nitrogen abundance is quite different for this 3 $M_{\odot}$ model, but we see differences in the other isotopes too. Especially the neon abundance is much larger in this model.

Carbon seems *less* abundant in the heavier model.

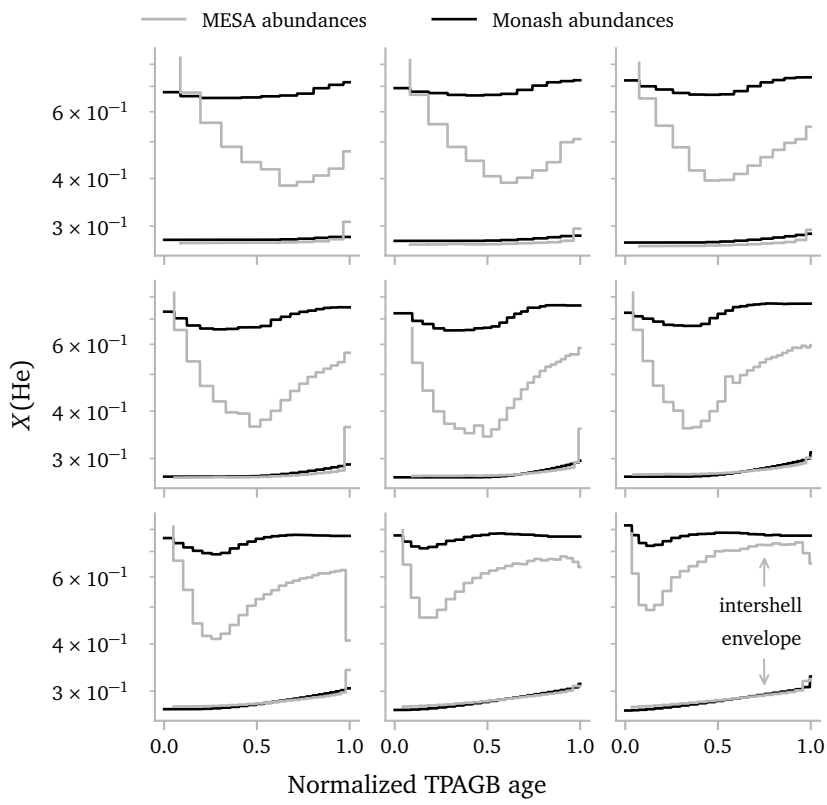
This figure below compares the MESA intershell abundances with the Monash post-processed intershell abundances for the 2.8 $M_{\odot}$ model.



The nitrogen abundance is very different between the two codes. The magnesium abundances initially are a bit different, but end up at nearly the same value.

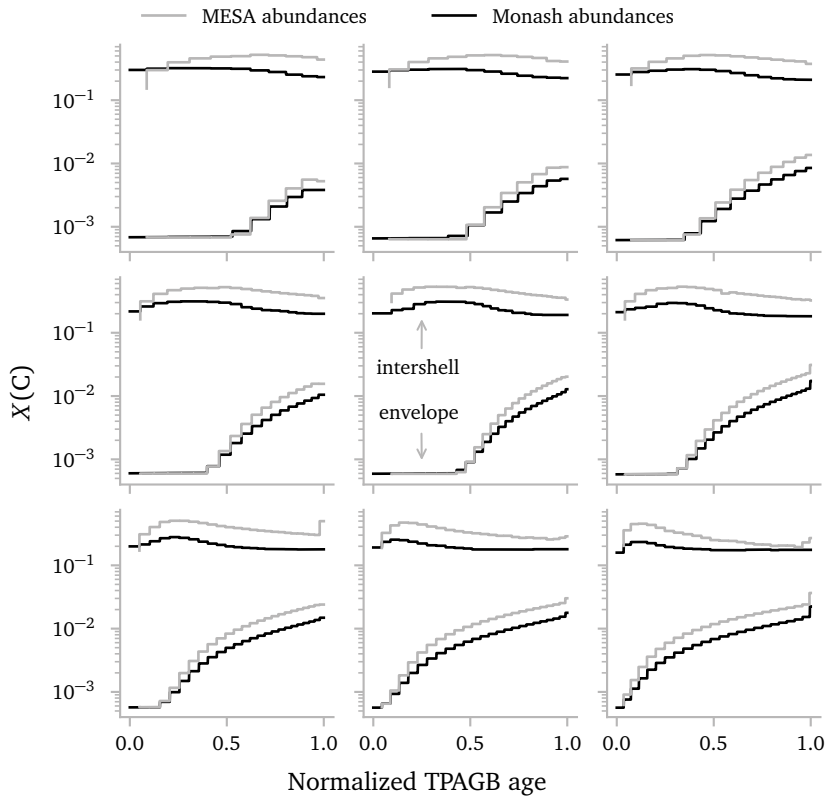
b.2 Envelope abundances

Now let me take a closer look at the intershell and envelope abundances for specific elements.



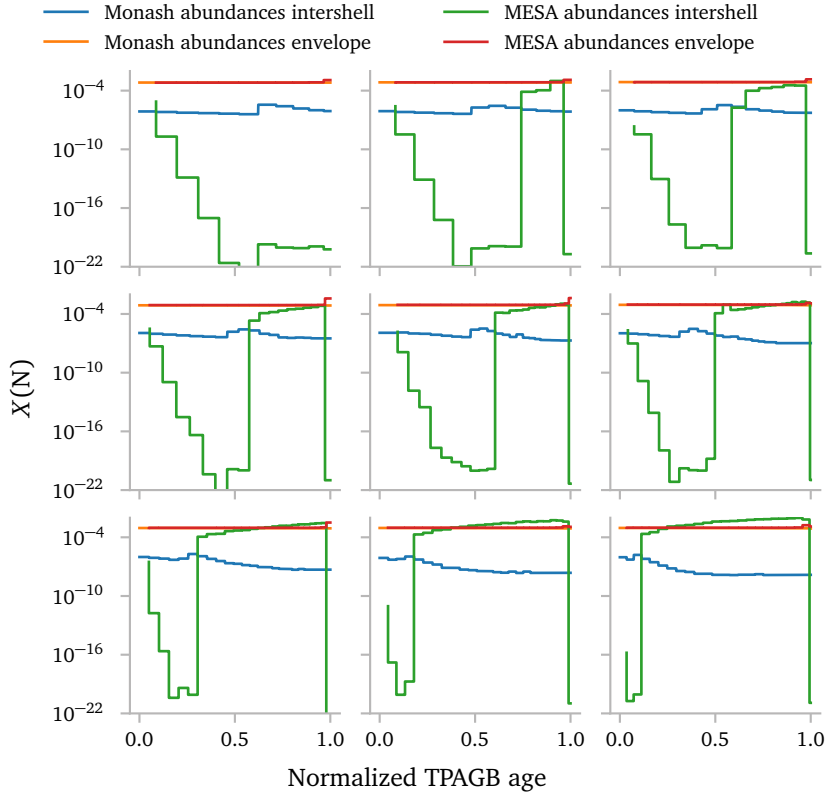
$M_{\text{TPAGB}} =$
1.4 1.6 1.8
2.0 2.2 2.4
2.6 2.8 3.0 $M_{\odot}$

The helium envelope abundances agree very well. Because of the new initial envelope abundance method, the initial values are very similar and because the intershell abundances are not much larger than the envelope abundances, dredge-up does not significantly impact the envelope abundances.



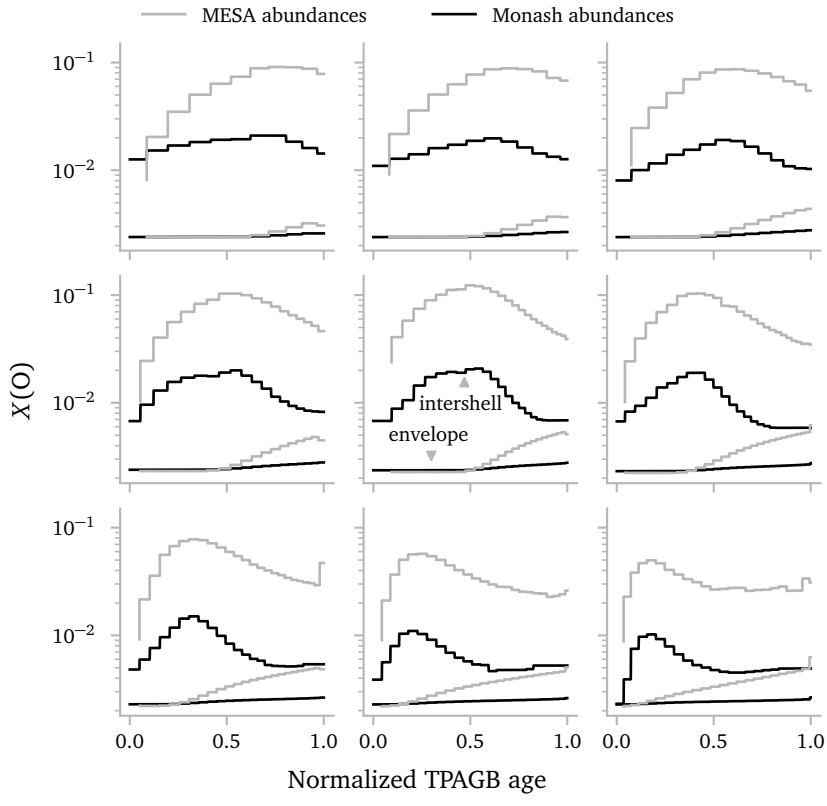
$M_{\text{TPAGB}} =$
1.4 1.6 1.8
2.0 2.2 2.4
2.6 2.8 3.0 $M_{\odot}$

The carbon abundances are a lot more interesting. The initial abundances are very similar. Just like last week however, we see that the carbon abundances in the envelope are lower for the Monash post-processed abundances. It makes sense because the intershell abundances are also lower for the Monash abundances.



$M_{\text{TPAGB}} =$
1.4 1.6 1.8
2.0 2.2 2.4
2.6 2.8 3.0 $M_{\odot}$

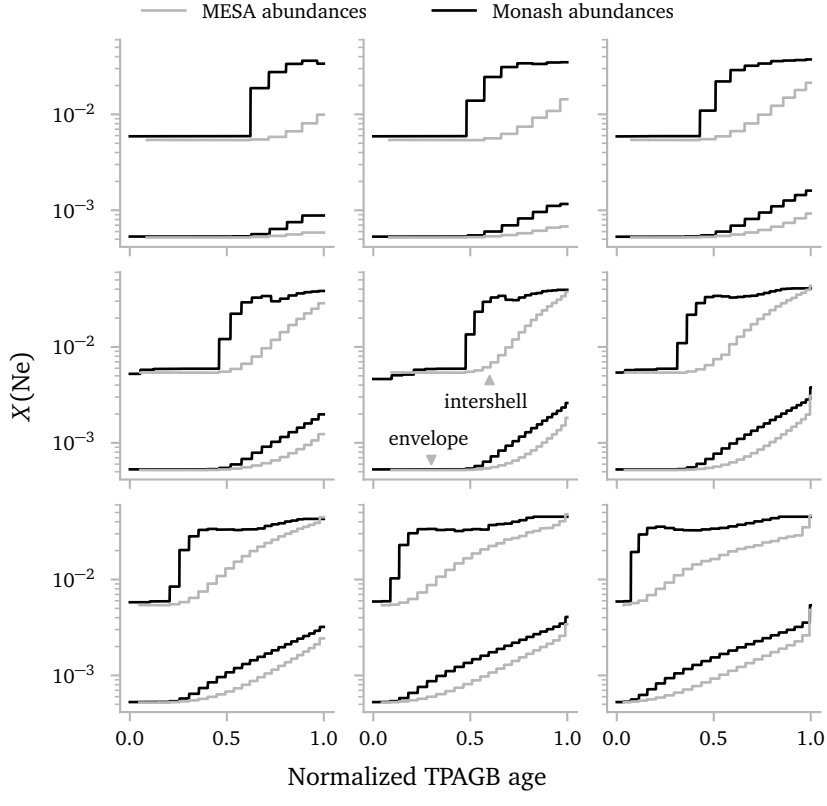
The intershell values are very different for the MESA and Monash abundances. I do not know why they are so different?



$M_{\text{TPAGB}} =$
1.4 1.6 1.8
2.0 2.2 2.4
2.6 2.8 3.0 $M_{\odot}$

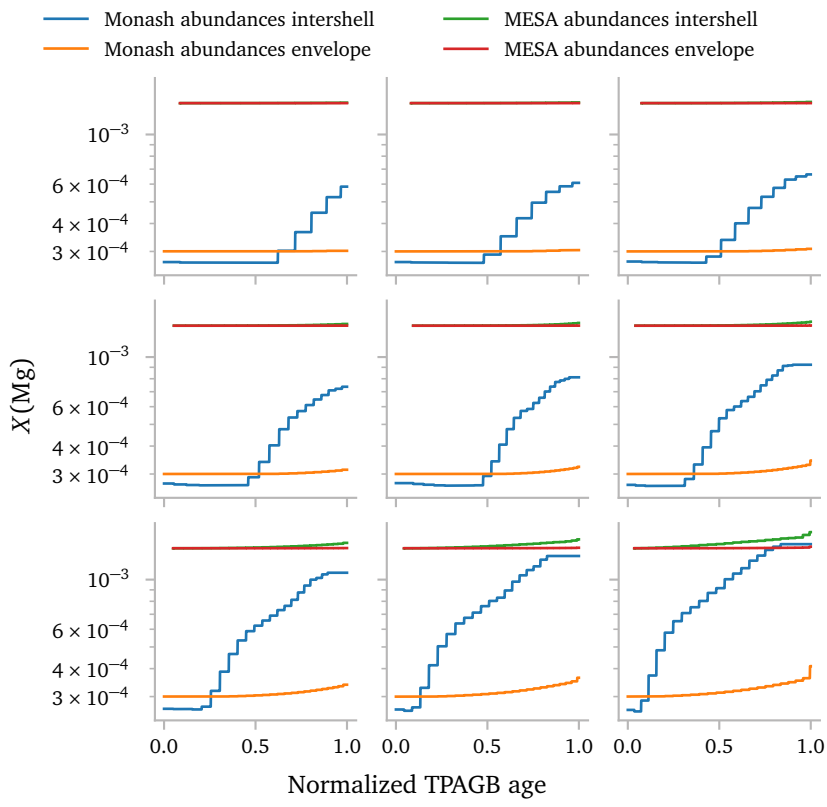
The initial oxygen envelope abundances once again agree very well. However, the intershell abundances are very different between the two codes. Because the Monash intershell abundances are much lower, the Monash post-processed envelope abundances do not increase with dredge-up as much as the MESA abundances.

The question is: why are the MESA intershell abundances so different from the Monash intershell abundances?



$M_{\text{TPAGB}} =$
1.4 1.6 1.8
2.0 2.2 2.4
2.6 2.8 3.0 $M_{\odot}$

I think the neon abundances are not very interesting. Overall both the envelope and intershell abundances agree pretty well. Just the shape of the Monash intershell abundance profiles rise earlier and more sharply which makes the Monash envelope abundances increase slightly more.

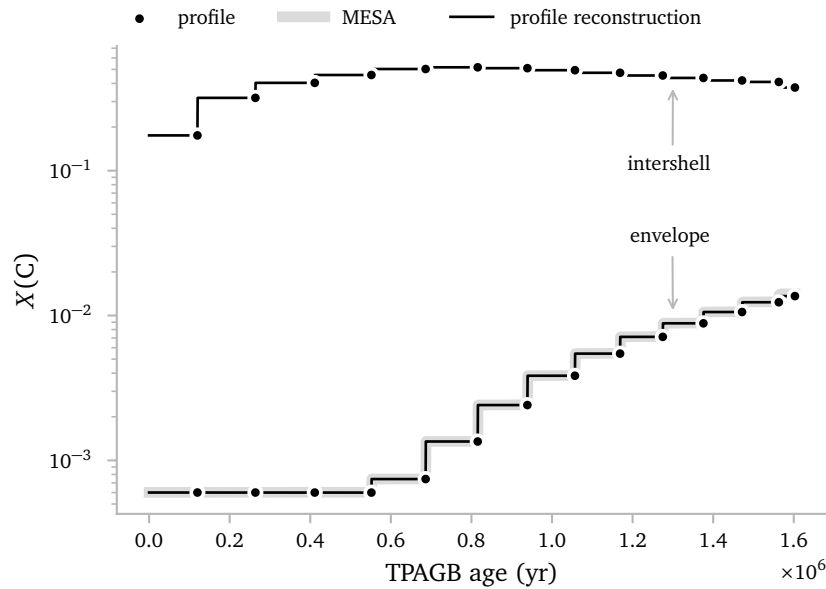


$M_{\text{TPAGB}} =$
1.4 1.6 1.8
2.0 2.2 2.4
2.6 2.8 3.0 $M_{\odot}$

The magnesium abundances are not very interesting because the initial MESA abundances are so much larger than the Monash initial abundances.

b.3 Post-processor with MESA abundances

Lastly, I want to test if my post-processing code reproduces the MESA envelope abundances if we supply it with the initial MESA envelope abundances with the MESA intershell abundances.

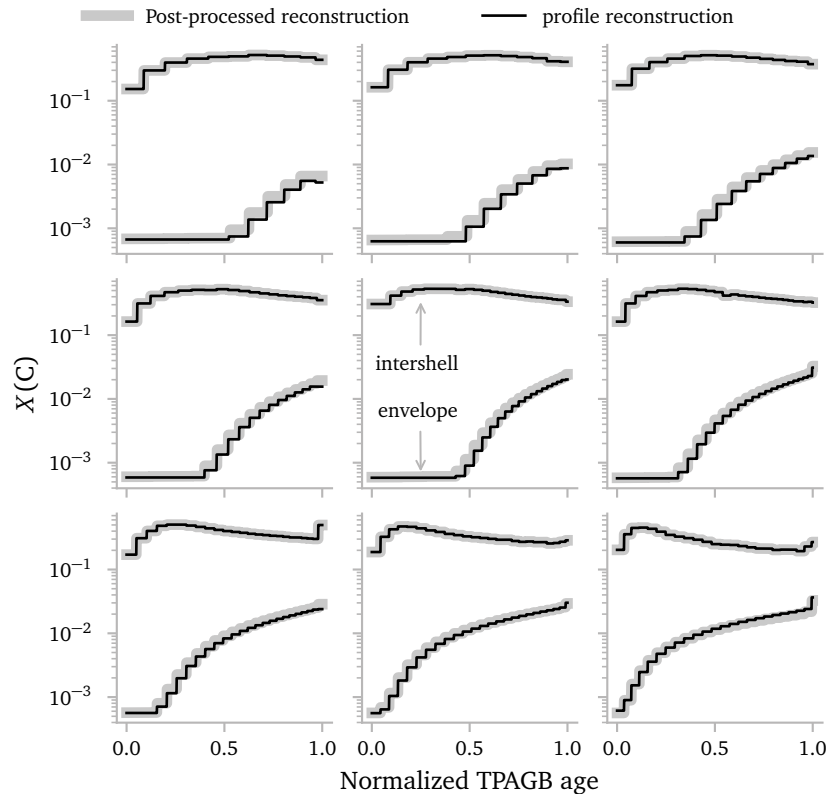


Firstly, I need to accurately reproduce the intershell abundance as a function of star model number. Now, since I only have the profiles, I need to make sure i correctly relate the sparse intershell abundances to the very densely spaces model numbers because my post-processing code expects this format.

I test that I correctly reproduce the envelope abundances.

Then I also produce the intershell values in the same way.

Now, let me give my post-processing code these intershell values and the initial envelope abundance.



$$M_{\text{TPAGB}} = \begin{matrix} 1.4 & 1.6 & 1.8 \\ 2.0 & 2.2 & 2.4 \\ 2.6 & 2.8 & 3.0 M_{\odot} \end{matrix}$$

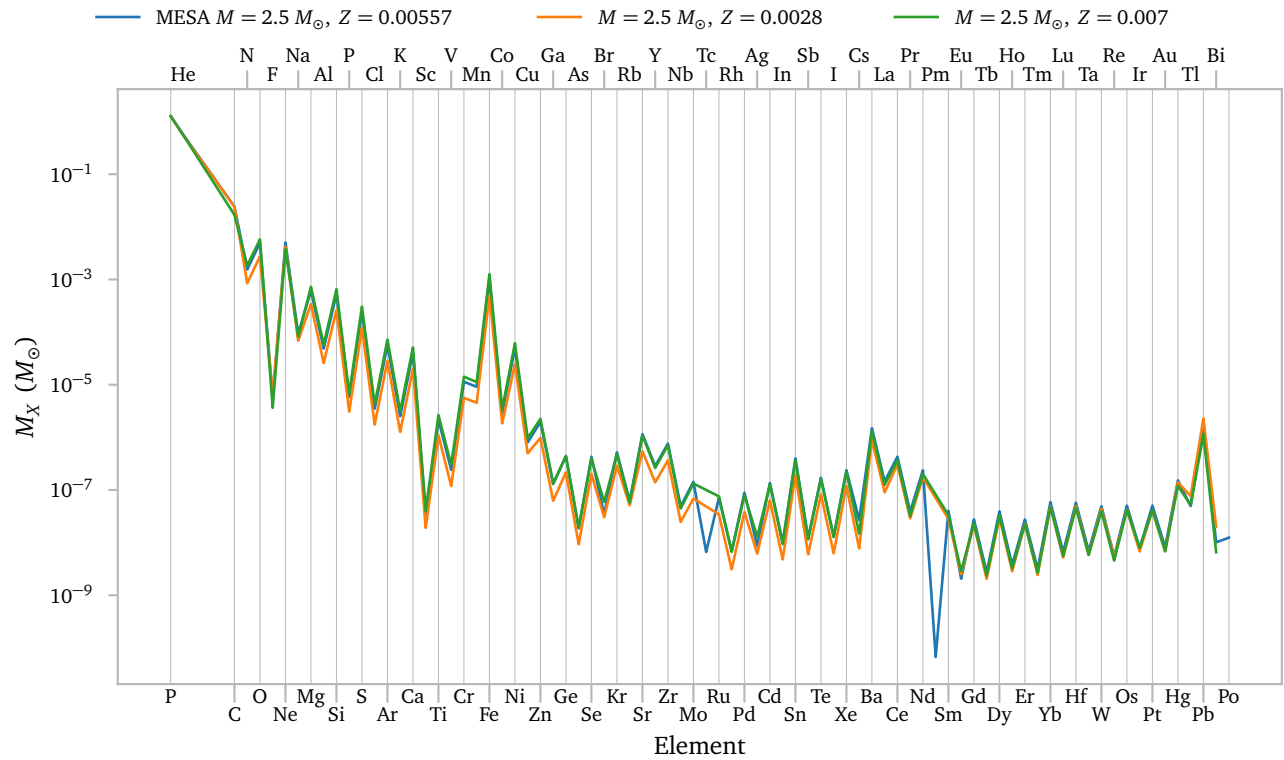
Clearly, the envelope value, *which* the post-processing code tries to reproduce, matches very well. This convinces me that my post-processor is correctly enriching the “fake” envelope.

This also means that the differences we saw in the plots above are only due to the differences in initial envelope abundance and intershell abundance evolution.

c. Investigating Monash Yields

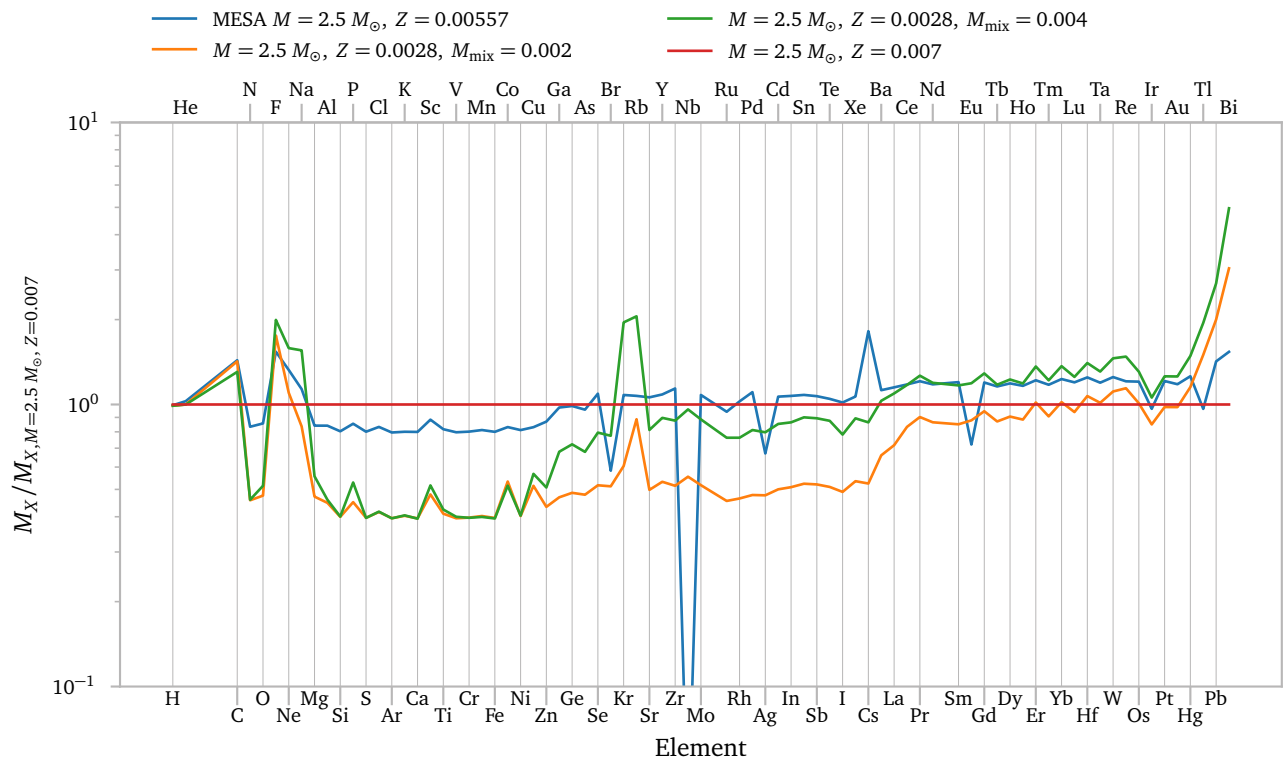
First things first, I made an error last week which quite drastically changed the results for the yields. However, the other results where all unaffected. The problem was in my implementation of the post-processor when it just uses the single star model. In this case, the post-processor does not have the parameters dict that the binary models do contain and the post-processor has to infer some of the relevant information itself.

Most importantly, it has to find the mass and the metallicity of the star. Now, I thought I would simply use the star property information for this, *but* apparently I had hardcoded a solar metallicity into the star model, which caused the post-processor to wrongly identify the models as solar metallicity models. It then used this metallicity to compute the interpolation and obtain the initial envelope abundances.



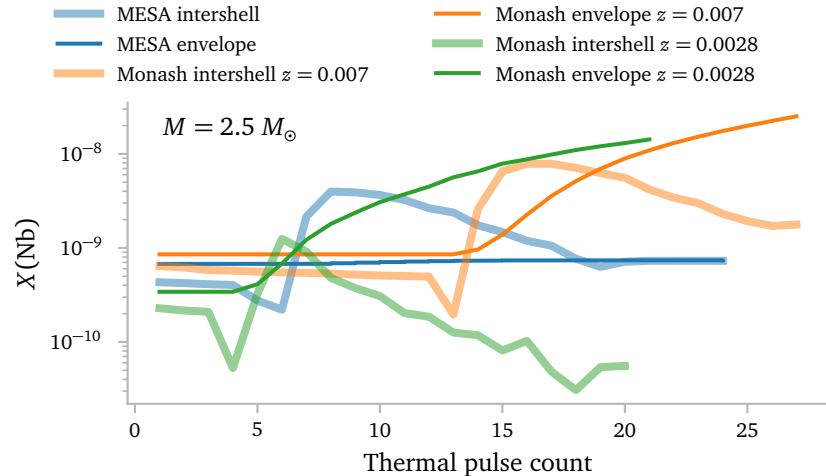
Let's also look at the differences.

NOTE: It is important to note that I also plot the $M_{\text{mix}} = 0.004 M_{\odot}$ model here. The MESA abundances are interpolated with the $M_{\text{mix}} = 0.002 M_{\odot}$ abundances, which is the standard value of the mixing parameter according to [Karakas & Lugaro \(2016\)](#). Thus, the *orange* line is much more relevant than the green one.



Compared to the plots from last week, this is *clearly* a much better match, and the offset is now actually much more in line with differences between the Monash models.

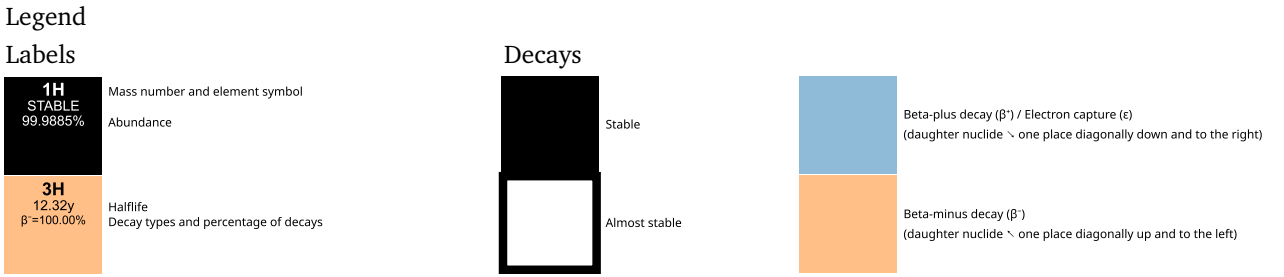
There are however a few isotopes with big differences in the abundances between MESA and the different Monash models. One example is Niobium, where we see a very big drop in the yield for the MESA models.



It seems like the envelope abundances of the Monash models are not *only* enriched by dredge-up. If this would have been the case, we would see a much straighter line for the envelope abundance, line in the MESA post-processed models. I think there must be some additional reactions that take place in the envelope after dredge-up.

Below, I plot part of the chart of nuclides adapted from [Wikimedia Commons \(2022\)](#).

91Ru 8.65 t=100.00%	92Ru 9.65m t=100.00%	93Ru 59.7s t=100.00%	94Ru 51.8m t=100.00%	95Ru 1.642s t=100.00%	96Ru STABLE 5.54%	97Ru 2.59s t=100.00%	98Ru STABLE 1.87%	99Ru 21.62s t=100.00%	100Ru STABLE 12.60%	101Ru 5.74s t=100.00%	102Ru STABLE 31.55%	103Ru 39.241s t=100.00%
90Tc 8.7s t=100.00%	91Tc 3.14m t=100.00%	92Tc 4.25m t=100.00%	93Tc 2.75h t=100.00%	94Tc 293m t=100.00%	95Tc 20.0h t=100.00%	96Tc 4.28d t=100.00%	97Tc 4.21·10 ⁶ y t=100.00%	98Tc 4.2·10 ⁶ y t=100.00%	99Tc 2.111·10 ⁵ y t=100.00%	100Tc 15.46s t=100.00%	101Tc 14.02m t=100.00%	102Tc 5.28s t=100.00%
89Mo 2.17m t=100.00%	90Mo 5.56m t=100.00%	91Mo 15.49m t=100.00%	92Mo STABLE 14.53%	93Mo 3.47·10 ⁶ y t=100.00%	94Mo STABLE 9.15%	95Mo STABLE 15.84%	96Mo STABLE 16.67%	97Mo STABLE 9.60%	98Mo STABLE 24.39%	99Mo 65.976h t=100.00%	100Mo 7.3·10 ⁶ y t=100.00%	101Mo 14.61m t=100.00%
88Nb 14.52m t=100.00%	89Nb 2.08h t=100.00%	90Nb 14.6m t=100.00%	91Nb 6.8·10 ⁶ y t=100.00%	92Nb 34.731d t=100.00%	93Nb STABLE 100%	94Nb 34.991d t=100.00%	95Nb STABLE t=100.00%	96Nb 23.25h t=100.00%	97Nb 72.11h t=100.00%	98Nb 2.86s t=100.00%	99Nb 15.0s t=100.00%	100Nb 1.5s t=100.00%
87Zr 1.68h t=100.00%	88Zr 83.4d t=100.00%	89Zr 78.41h t=100.00%	90Zr STABLE t=100.00%	91Zr STABLE 11.22%	92Zr STABLE t=100.00%	93Zr 1.61·10 ⁶ y t=100.00%	94Zr STABLE 17.38%	95Zr 64.032d t=100.00%	96Zr 2.35·10 ⁶ y t=100.00%	97Zr 16.749h t=100.00%	98Zr 30.7s t=100.00%	99Zr 2.1s t=100.00%
86Y 14.52m t=100.00%	87Y 79.0h t=100.00%	88Y 100.625d t=100.00%	89Y STABLE 100%	90Y 64.003h t=100.00%	91Y 58.51d t=100.00%	92Y 5.40h t=100.00%	93Y 10.18h t=100.00%	94Y 16.7m t=100.00%	95Y 10.3m t=100.00%	96Y 5.34s t=100.00%	97Y 5.75s t=100.00%	98Y 0.548s t=100.00%
85Sr 64.849d t=100.00%	86Sr STABLE 9.86%	87Sr STABLE 7.03%	88Sr STABLE 82.58%	89Sr 50.563d t=100.00%	90Sr 28.90y t=100.00%	91Sr 9.65h t=100.00%	92Sr 2.66h t=100.00%	93Sr 7.43m t=100.00%	94Sr 75.3s t=100.00%	95Sr 23.90s t=100.00%	96Sr 1.07s t=100.00%	97Sr 429ms t=100.00%

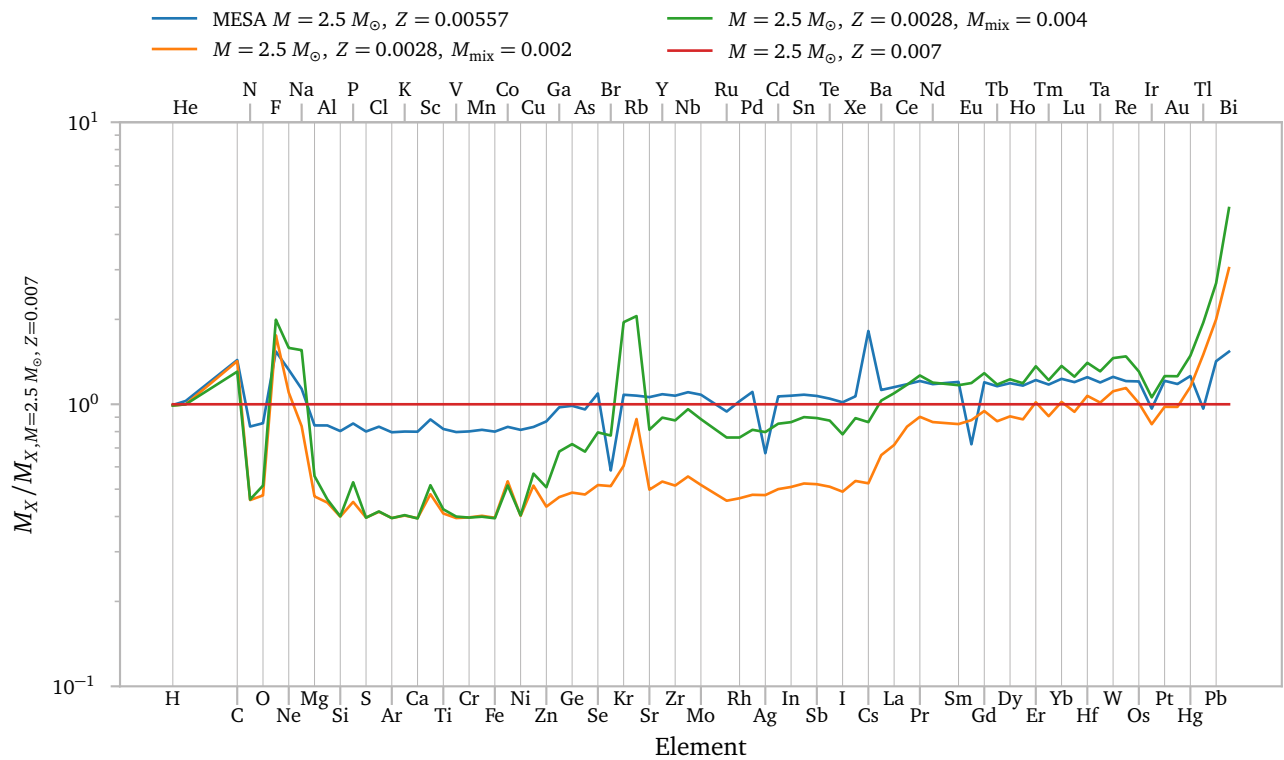
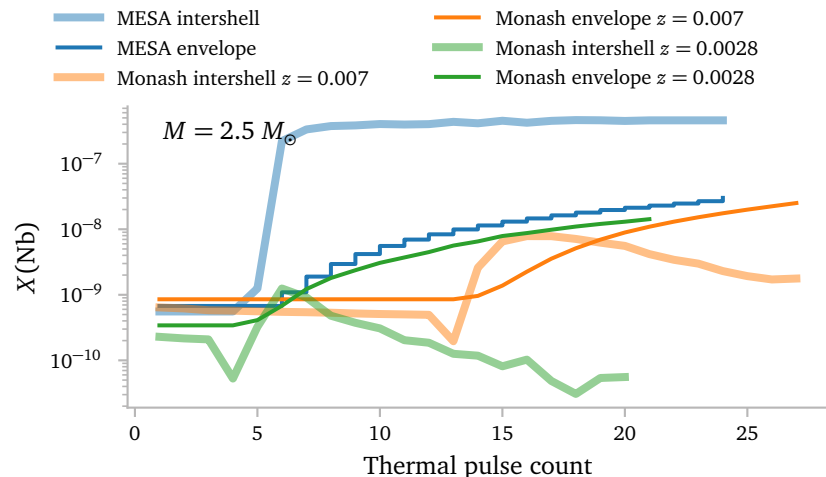


I think the interesting isotope here might be ^{93}Zr , which has a long half-life, but it decays to a stable isotope of our underrepresented element. It is a bit difficult to look at the *envelope* abundances of ^{93}Zr , because we only have information on the *element* Zr, not the individual isotopes.

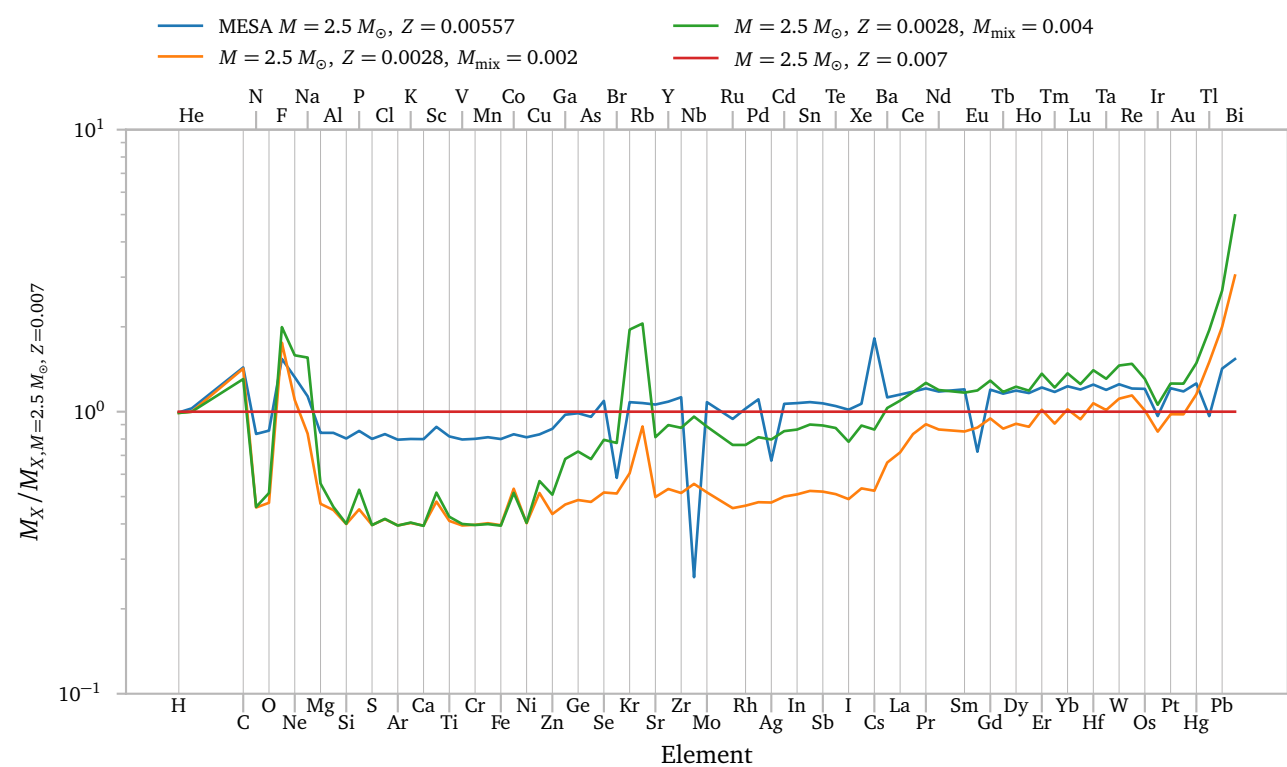
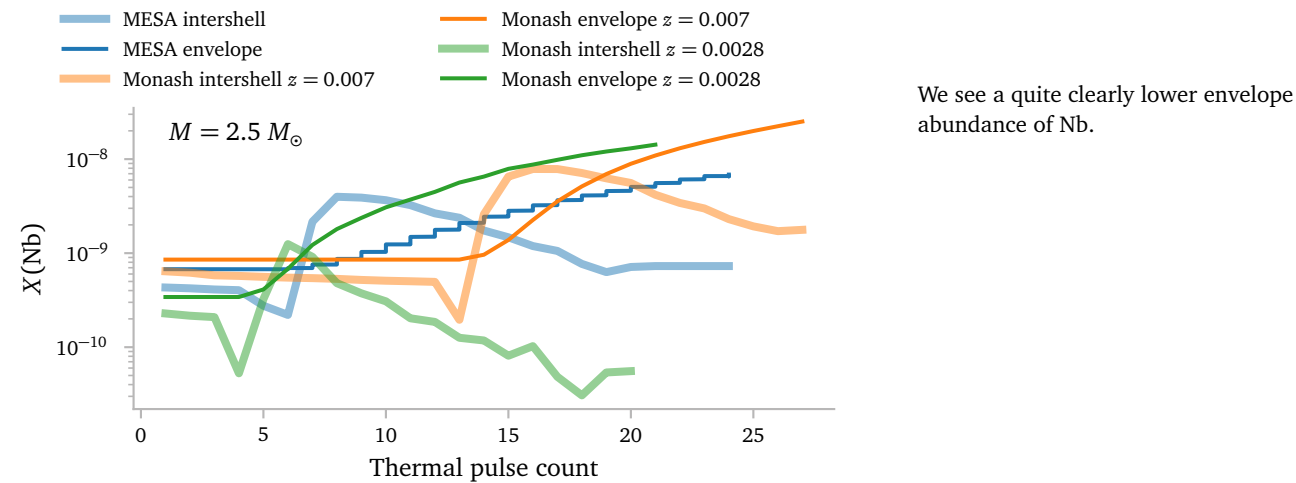
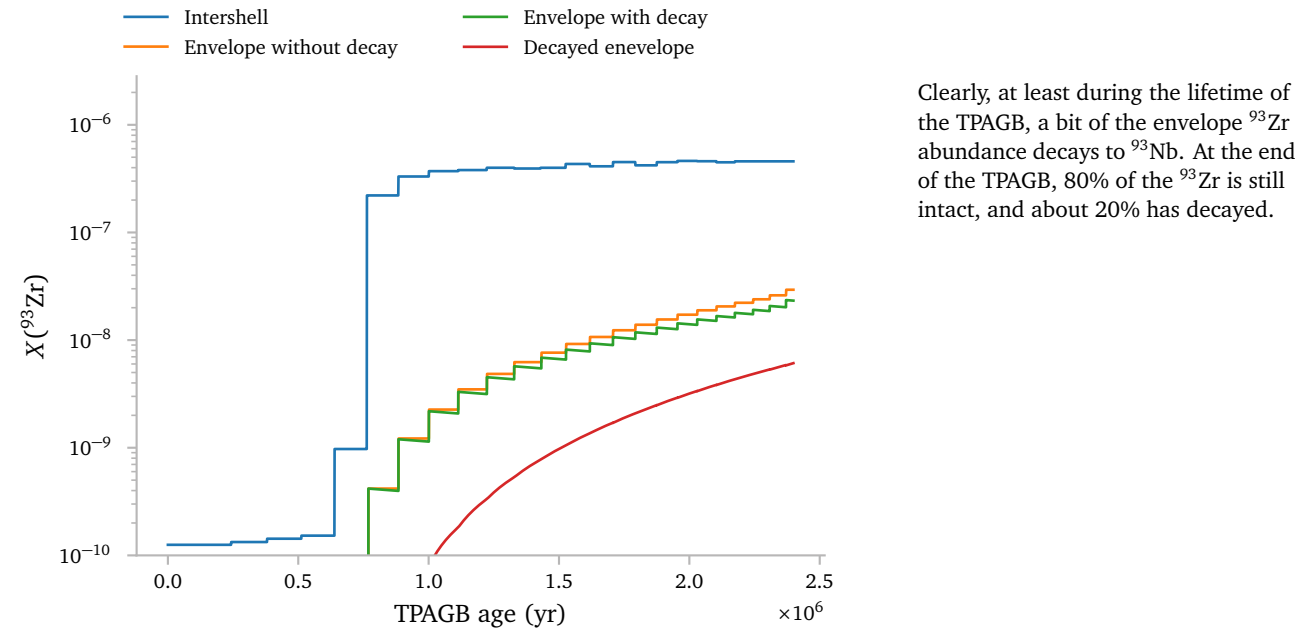
In the paper from [Karakas & Lugaro \(2016\)](#), they write the following about ^{93}Zr :

“ Note that we do not decay the abundances of radioactive isotopes (e.g., ^{26}Al , ^{60}Fe) in the isotopic or elemental surface abundance files, but they are assumed to have all decayed in the yield tables. However, we do decay the isotope ^{93}Zr to ^{93}Nb because Nb obtains essentially all of its production via this decay. ”
Karakas & Lugaro (2016)

I think I can do the same, but I am not quite sure why they do this. The half-life time is about 1.6 million years and this, at least to the same order of magnitude, about the same duration of the TPAGB phase of stars with masses between $1.5 M_{\odot}$ and $3.0 M_{\odot}$.



This looks quite good and I think this is what [Karakas & Lugaro \(2016\)](#) did as well. It makes sense since we would expect that, even if the material is expelled as ^{93}Zr , it will eventually become ^{93}Nb . However, I am curious about actually simulating the decay *in the envelope*. For this, I assume that ^{93}Zr has *no* initial abundance in the envelope, since it is an unstable isotope.



The yield fits a lot worse this time. With the previous method, I would say that the differences between the individual Monash models are bigger than the differences between the *MESA* model and the reference Monash model, which means we are probably at a good spot.

There are however a few other elements that are still a bit “weird”. With the exception of Cs, these are all elements that quite significantly underrepresented in the *MESA* yields. I think here the story is similar to the story of Nb. There is some isotope that decays to the underrepresented element, probably with quite a long decay time.

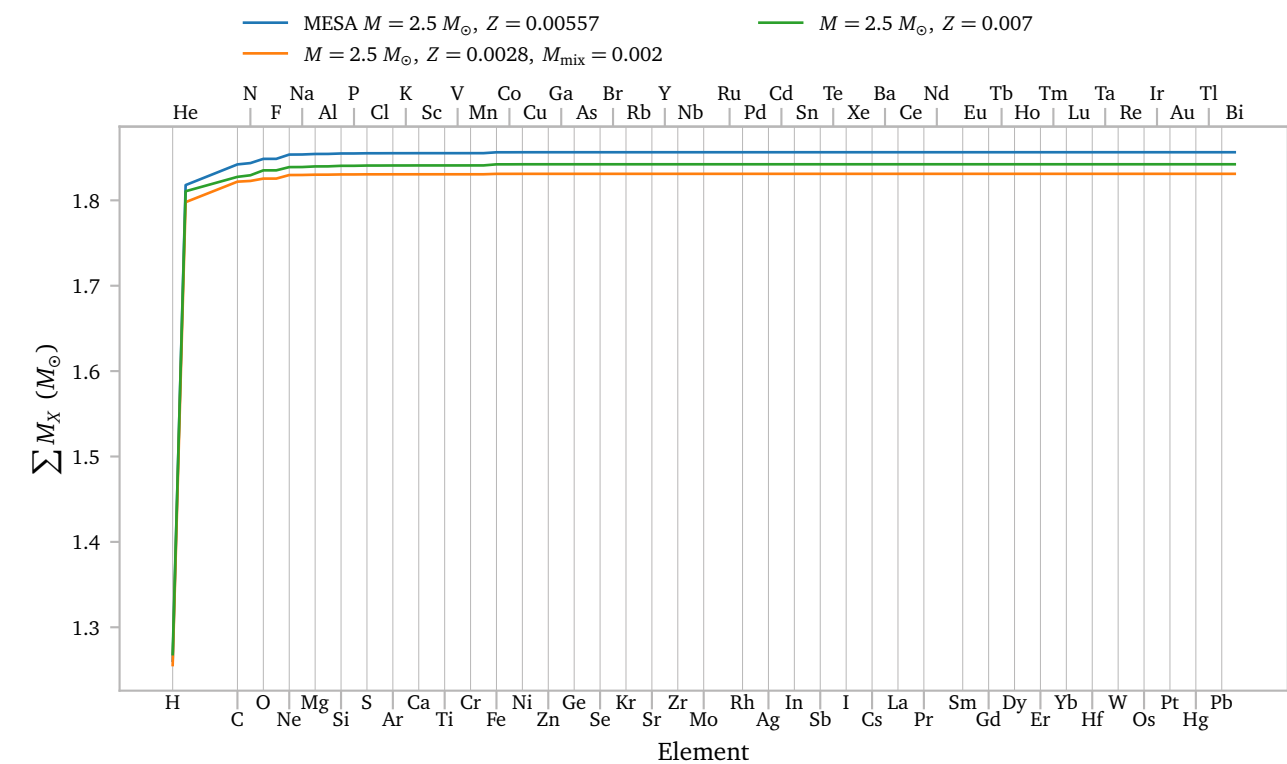
For these isotopes, I could implement a similar procedure. This however requires a lot of manual work, or quite a bit of setup work to load some nuclides table, compare every isotope with this decay process in the table and implement the correct decay.

I think my interpolation probably introduces more uncertainty than the reactions that happen in the envelope or in the interstellar medium after the mass has been expelled.

However, this problem is obviously not only there for the yields, but also for the envelope abundances as a function of time, which is easy to see if we look at the ^{93}Zr abundance plot above.

The last thing I want to investigate is the effects of the total expelled mass.

Naively, one would expect that models with the same initial mass would expel roughly the same amount of mass. *However*, differences in the core mass do change the total amount a little bit.



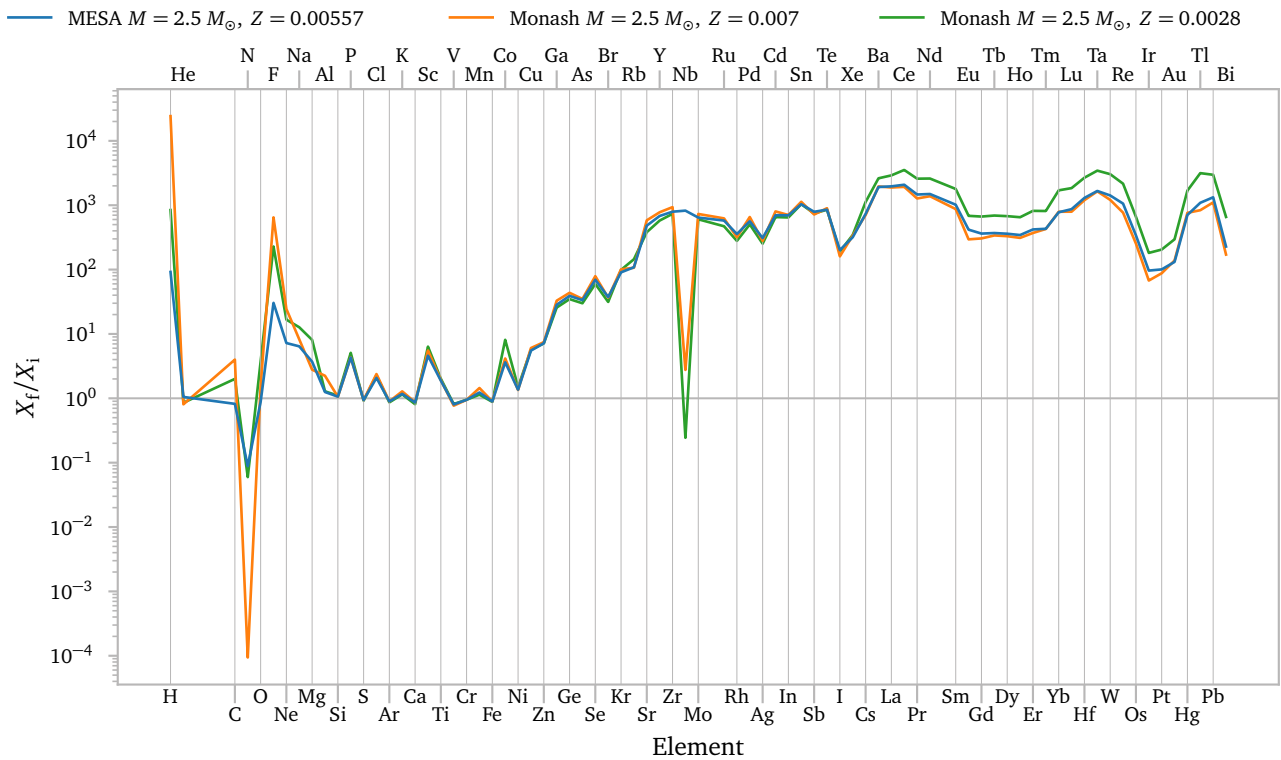
These differences however are so small that after “correcting” for this, the abundances distributions changes are minimal.

We see that the especially for s-process elements, the *MESA* model predicts slightly *higher* yields than both the Monash models. I think this makes sense if we take into account the fact that the *MESA* models often experience the first-dredge up *earlier*. This means the stars will have more thermal pulses to dredge-up enriched intershell material to the envelope. If the star has higher envelope abundances at the end of the evolution when mass losses become important, we will obviously see *larger* yields for these elements.

d. Envelope and Intershell Abundance Changes

d.1 Intershell

Below, I show the fraction of the final intershell abundance over the initial intershell abundance.

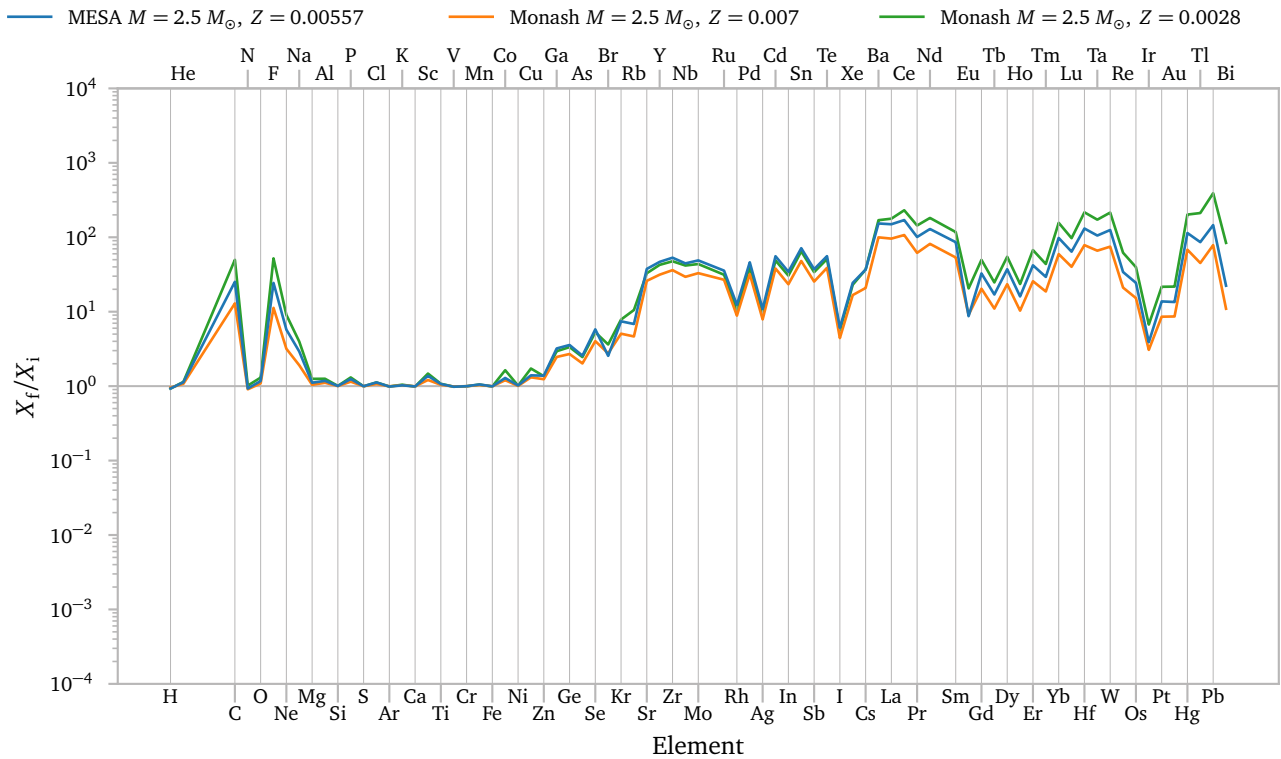


I think this quite clearly matches *very* well for the *MESA* models and the Monash models, which is unsurprising because the MESA intershell abundances should just be the interpolated between the two shown Monash intershell values. The only reason they *could* deviate a lot is if for some reason, the MESA model *with* offset, does not manage to reach the last thermal pulse index in each of the Monash models. In this case, the final MESA abundances will correspond to the abundances of a *not* final Monash model pulse. This still would only change the intershell abundances a lot if for some reason, the Monash abundances change a lot after already a lot of dredge up occurred, which I cannot imagine being the case.

We see that the MESA models generally follow the $Z = 0.007$ models better than the $Z = 0.0028$ models.

d.2 Envelope

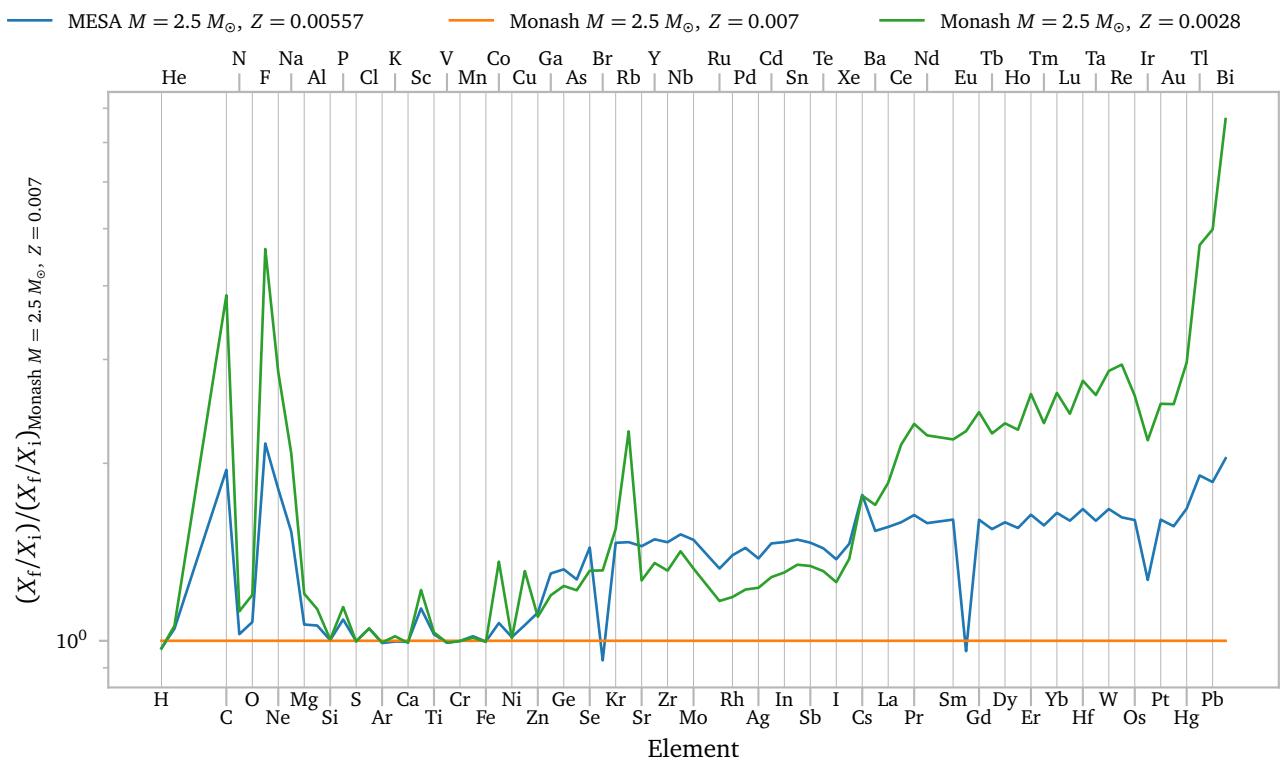
Now, obviously we can make the same plot for the *envelope*. In this case, we *expect* the MESA model abundances to become slightly higher than the $Z = 0.007$ models, simply because the star will dredge up more of the intershell abundances, which will generally be much more abundant in heavy elements.



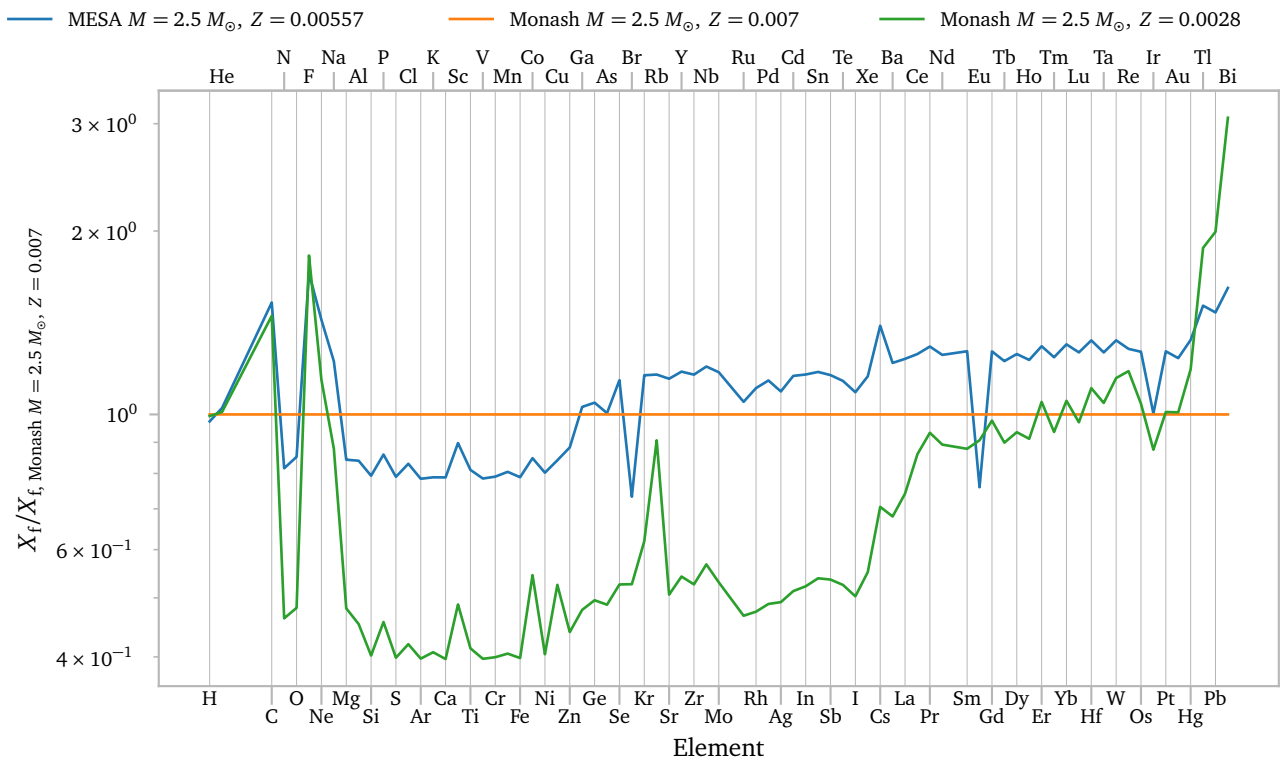
This is exactly what we see, because the MESA models now much closer to the $Z = 0.0028$ models.

Also, for basically all elements between magnesium and iron, nothing changes between in the envelope due to dredge-up.

Let me now just divide the lines we see in the plot above. This way we see more clearly the ratio between them.



Now lastly, we can just look at the final envelope abundances. In this case, I compare it to a reference model again.



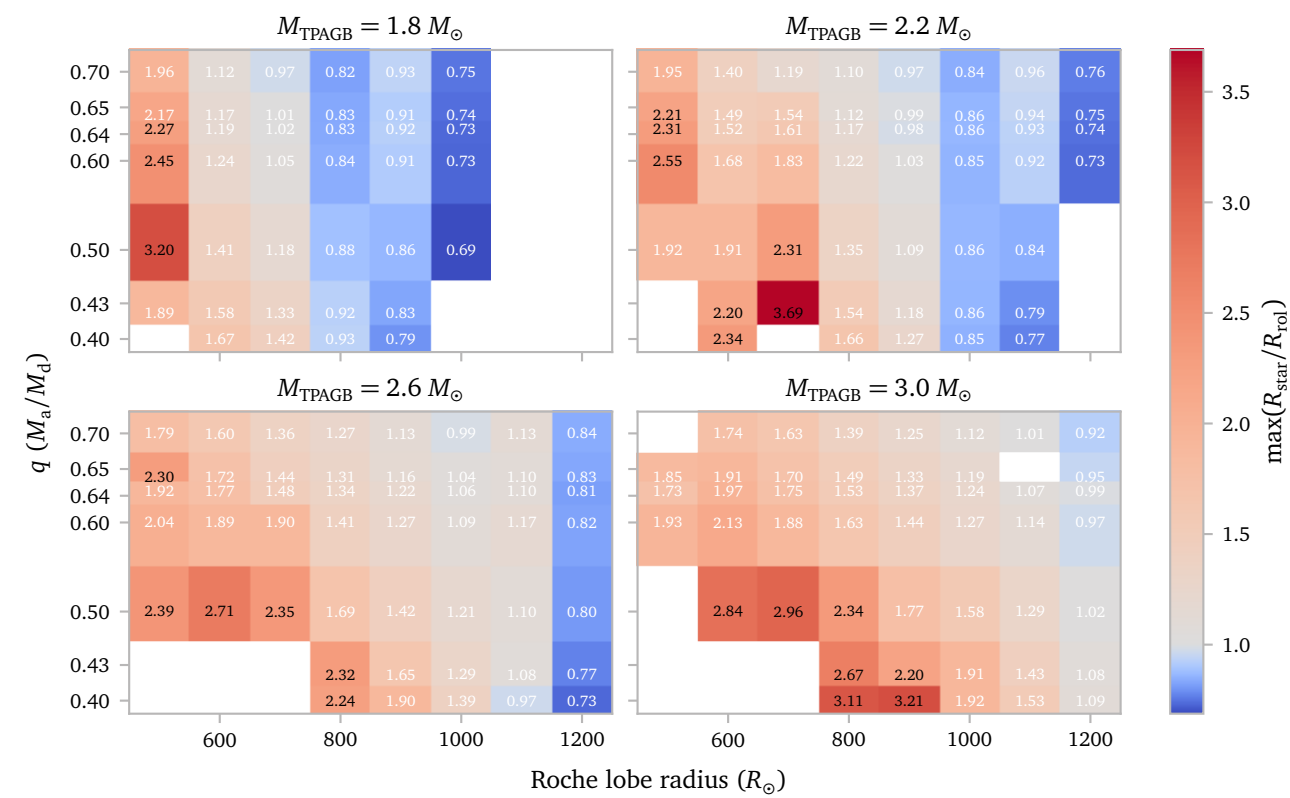
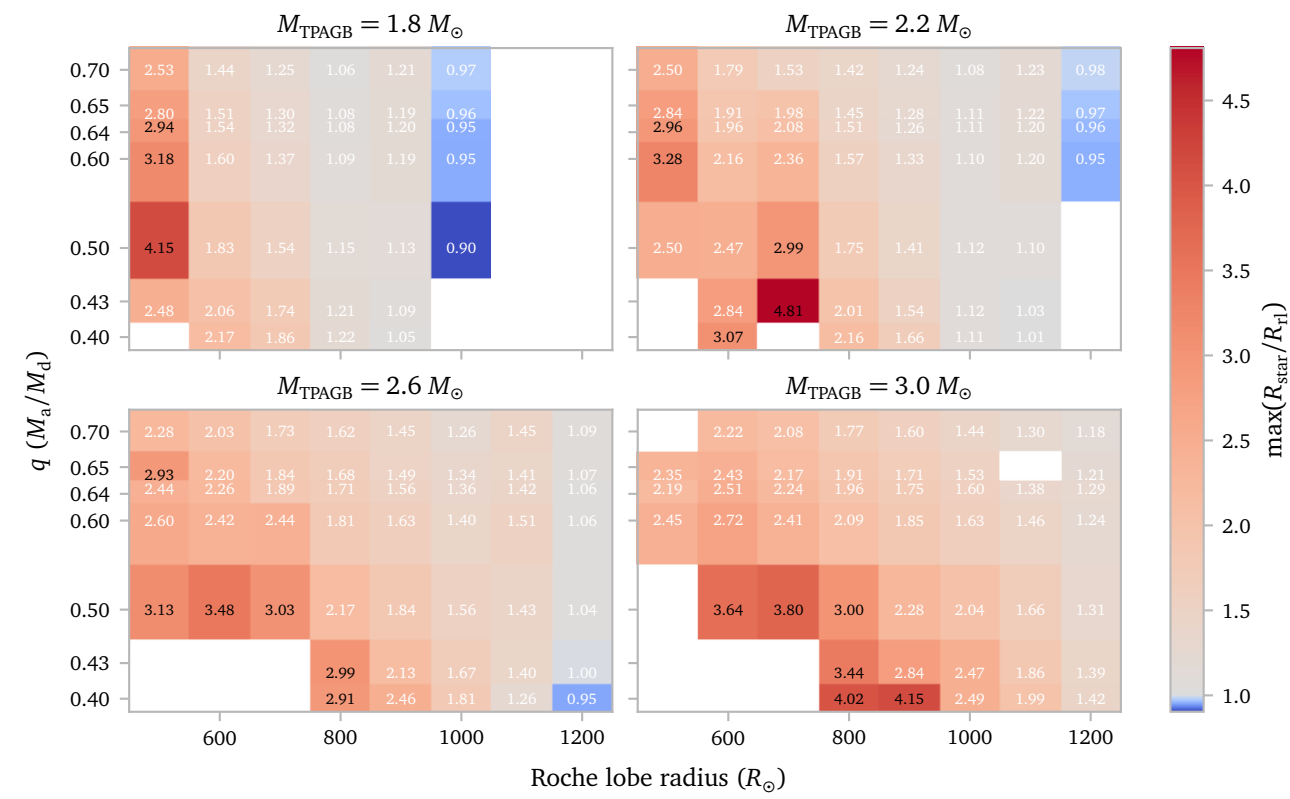
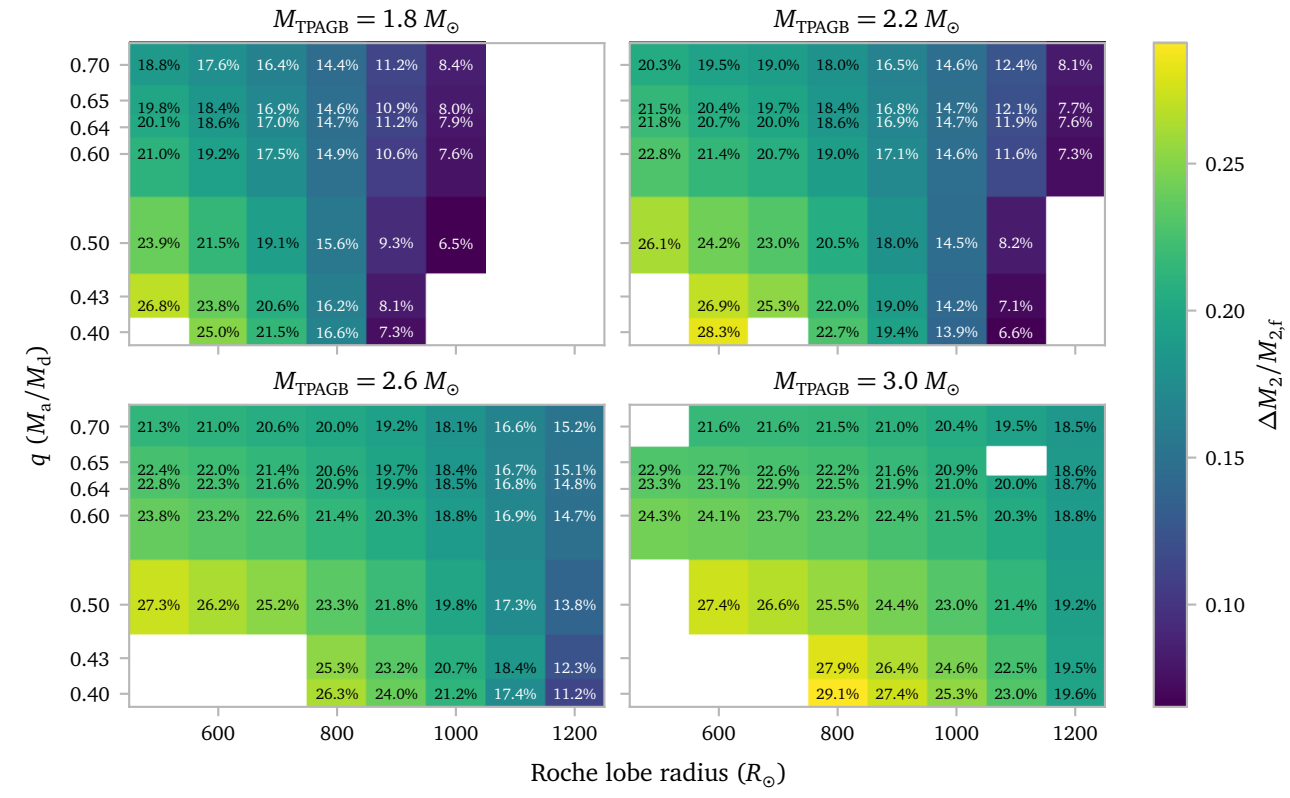
We obtain the same shape as we do for the yields.

Because we already saw that the elements magnesium through iron are not enriched, these elements only have comparably *lower* final mass fractions in the envelope because of their initially *lower* mass fraction purely due to the lower metallicity.

We also saw that dredge-up is more powerfull for metal poor Monash models, and my MESA models, which together with the intershell enrichment, causes the final envelope abundances of s-process elements to become *larger* for the MESA models and even the lower metallicity Monash models.

I feel like I now understand the shape of the yield and envelope abundance plots quite well. The differences between the models for individual models require looking into specific s-process reactions, which I have not done except for Nb.

e. Accretion and Mixing in MS stars



References

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